QSCP Corporate Asset Integrity

Corporate Summary

No.	Status	Date
QSCP-CI-SUM-2025-01	Issued	2025-05-05

Fleet Integrity Summary - Rich Amine Thinning

Operational History and Damage Mechanism

Following QSCP-MOC-2014-017, which increased DEA strength from 25 to 30 wt% to accommodate higher sour-gas throughput in ATU-100, a corresponding QSCP-DMR-2014-002 identified an increased risk of rich-amine thinning. The review anticipated higher acid-gas loading would accelerate corrosion rates, necessitating a 3-year inspection interval and additional monitoring downstream of the let-down valve. Further acceleration of metal loss was observed following a let-down valve trim change in 2019.

Inspection Findings and Nonconformance

During the ATU-100 five-year survey documented in LNDE-RPT-20241025-031-R2, ultrasonic thickness measurements on line 4-RA-101 (A106 Gr.B, 114.3 mm OD, 8.56 mm t_nom) revealed significant wall loss. Specifically, QSCP-NCR-2024-031 was issued after CML 4-RA-101-06 was measured at 3.59 mm on 2024-11-04, falling below the 3.81 mm t-min. A follow-up reading on 2024-12-10 confirmed a thickness of 3.55 mm, resulting in an expired remaining life based on a calculated corrosion rate of 0.659 mm/y.

The thinning trend at CML 4-RA-101-06 shows a sharp increase from 0.029 mm/y in 2016 to 0.664 mm/y in late 2024. This document, authored by Dr. Helen Marchetti, summarizes the integrity status for the fleet based on these findings.

Sign-off	Name	Function	Date	Signature
Prepared	Dr. Helen Marchetti	Corporate Asset Integrity Manager, P.Eng.	2025-05-05	_____